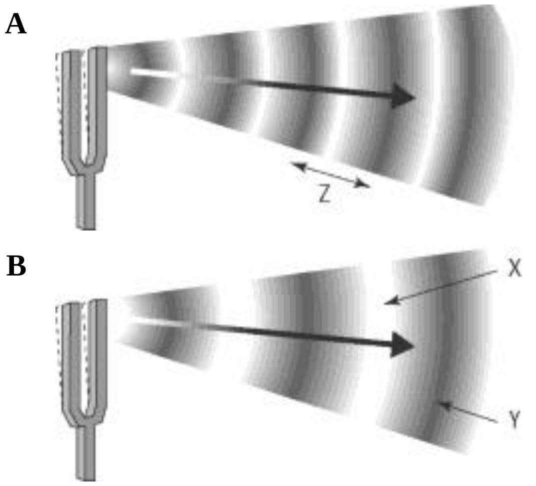
Worksheet 9.1

Sound energy

Answers

Wavelength and frequency

1. The following diagram shows two vibrating tuning forks producing sound waves.



(a) Identify the features of the wave labelled X, Y and Z.

X: rarefaction

Y: compression

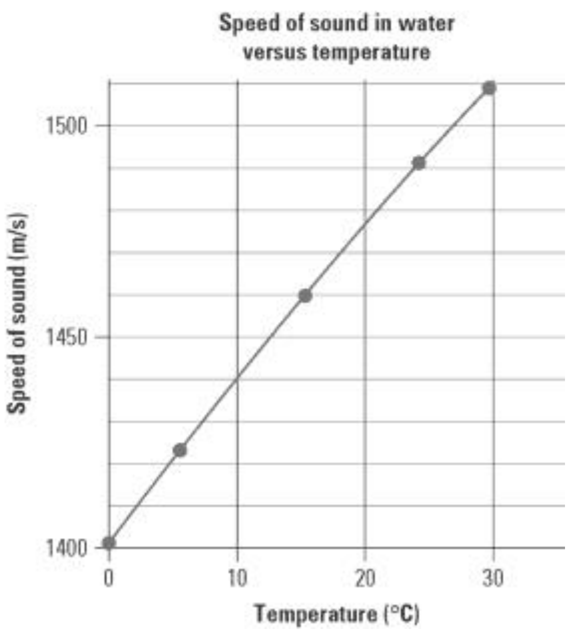
Z: wavelength

(b) Which sound wave has the lowest pitch? Explain.

B has the lowest pitch (or frequency) as the wavelength is greater.

Speed of sound in water

2. The following graph shows how the speed of sound (in m/s) in water varies with temperature (in °C).



(a) The labels and units on each axis are not shown. Insert these labels and units.

(b) Give the graph a title.

(c) Use the graph to determine the speed of sound in water at:

(i) 0 °C 1405 m/s

(ii) 10 °C 1440 m/s

(iii) 20 °C 1475 m/s

(d) In air, the speed of sound at 0 °C is 331 m/s.

(i) How many times faster does sound move in water than air at that temperature? (*Hint:* Divide one speed by the other.)

$\frac{1405}{331} = 4.20$. Over four times faster.

(ii) Explain why sound moves faster in water than in air.

The sound moves faster in water than air as the particles are much closer together and so collisions occur more frequently and thus the energy is passed on to the next particle more quickly.